EMOTIX: A WEARABLE AI SYSTEM FOR EMPLOYEE ASSISTANT AND MENTAL WELLNESS

25-26J-007

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
Specializing in Computer Systems & Network Engineering

Department of Computer Systems Engineering

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or

Diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Workplace stress and emotional well-being challenges have reached critical levels, necessitating proactive monitoring and intervention systems to support mental health in real-time environments. This research develops an intelligent stress detection system that utilizes facial expression analysis to identify stress-related emotions through machine learning and deep learning approaches. The system operates by capturing facial data through camera integration with edge devices, enabling continuous emotional state monitoring in natural environments.

Technical architecture employs hybrid CNN-LSTM models that combine convolutional neural networks with long short-term memory networks to enhance emotion detection accuracy. This approach captures both spatial facial features and temporal patterns, providing deeper understanding of stress progression beyond conventional detection methods. Lightweight machine learning models are implemented specifically for edge computing environments, ensuring efficient processing capabilities on local devices.

Upon detecting elevated stress levels, the system immediately transmits alerts and personalized suggestions to user devices via MQTT protocol. An intelligent recommendation engine selects the most appropriate interventions from a comprehensive pool of stress-relief techniques, including breathing exercises, calming music, mindfulness prompts, and break reminders. The selection process utilizes recommendation logic to match interventions with individual user contexts and preferences.

All system activities are comprehensively logged, including stress events, alert notifications, and user actions, with precise timestamp recording for detailed behavioral analysis. An interactive dashboard presents emotional trends and patterns, supporting ongoing emotional health monitoring and well-being assessment. This visualization component enables users to track their stress patterns and evaluate intervention effectiveness over time.

The research contributes a complete stress monitoring ecosystem that bridges the gap between emotion detection and practical intervention delivery. Expected outcomes include improved accuracy in stress identification, enhanced user engagement through personalized recommendations, and comprehensive emotional analytics for sustained mental health support. This work advances digital health technology by demonstrating effective integration of computer vision, edge computing, and personalized therapeutic interventions in accessible, privacy-focused applications.

Keywords**: Stress Detection, Facial Expression Recognition, CNN-LSTM Models, Edge Computing, Personalized Interventions, Emotional Analytics, Digital Health, Mental Well-being.**

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
CNN–LSTM	Hybrid Convolutional Neural Network and Long Short-Term Memory
IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
REST	Representational State Transfer
API	Application Programming Interface
UI/UX	User Interface / User Experience
PC	Personal Computer
DB	Database
GPU	Graphics Processing Unit
AES	Advanced Encryption Standard
TLS	Transport Layer Security
CI/CD	Continuous Integration / Continuous Deployment
SUS	System Usability Scale

1. INTRODUCTION

1.1 Background

Workplace stress has become a critical challenge in modern organizations, contributing to decreased productivity, increased healthcare costs, and widespread employee burnout. The World Health Organization reports that stress and related mental health conditions affect over 264 million people globally and cost the global economy more than $1 trillion annually in lost productivity [3]. Employees aged 30 to 40 years are especially vulnerable, facing mid-career pressures, heavier workloads, and the dual responsibility of managing professional and personal commitments. This makes them a crucial demographic for proactive stress detection and support systems.

Traditional stress management often relies on reactive interventions, where help is sought only after severe symptoms or crisis points [6]. Such reactive methods have proven inadequate for today's escalating workplace mental health crisis [15]. Artificial intelligence (AI) and computer vision technologies present opportunities to transform stress monitoring into a proactive, real-time process [4], [5]. Unlike self-reports or invasive physiological sensors, facial expression analysis offers continuous, non-intrusive, and objective insights into emotional well-being. Subtle facial expressions and micro-movements often reveal stress before individuals consciously recognize it [14]. Integrating these AI capabilities with edge computing further enhances privacy, accessibility, and speed. Real-time local analysis on edge devices prevents transmission of sensitive data to external servers, addressing privacy concerns while enabling immediate stress detection and rapid intervention [11].

The convergence of emotion recognition, personalized interventions, and continuous behavioral tracking establishes the foundation for a complete stress management ecosystem. Such systems detect emotional distress, deliver timely support including mindfulness prompts, breathing exercises, or relaxation reminders and maintain long-term monitoring of well-being. This represents a paradigm shift toward proactive mental health care, where employees receive real-time, personalized support that prevents stress from escalating into serious health or performance issues [7].

1.2 Literature Review

1.2.1 Traditional Stress Detection Methods

Traditional stress detection has largely relied on self-report questionnaires, periodic surveys, and physiological measurements. While valuable, these approaches require conscious user input and are prone to bias (e.g., social desirability, recall errors) and delayed recognition of symptoms [2], [6]. Tools such as the Perceived Stress Scale and Depression Anxiety Stress Scales depend on individuals accurately reporting their emotional state, usually after stress has already intensified. Physiological monitoring via heart rate variability, galvanic skin response, or cortisol levels shows promise in laboratory settings but faces challenges in real-world workplaces due to the

need for specialized sensors, invasive procedures, or impractical sample collection. Moreover, these physiological signals are indirect indicators of stress and can be significantly confounded by non-stress factors like physical activity, caffeine intake, or health conditions, leading to false positives. The reactive nature of these traditional methods means interventions often occur only after stress becomes severe, missing the opportunity for early prevention and support [6].

1.2.2 Convolutional Neural Networks in Facial Expression Recognition

Computer vision and deep learning have introduced new methods for stress detection, with Convolutional Neural Networks (CNNs) proving highly effective in facial expression recognition [4], [8]. CNNs automatically extract spatial features (such as facial edges, textures, and muscle action patterns) correlated with emotional states, enabling detection of subtle facial movements linked to stress, anxiety, frustration, or fatigue [9]. Their hierarchical, layered learning structure transforms raw pixel data into increasingly abstract representations of emotions, providing an objective and non-intrusive window into well-being [10]. However, standard CNN-based approaches typically analyze static images, limiting their ability to capture emotional transitions or stress accumulation over time. This temporal "blind spot" reduces their effectiveness in continuous monitoring scenarios, where understanding the progression of stress is crucial.

1.2.3 LSTM Networks for Temporal Pattern Analysis

Long Short-Term Memory (LSTM) networks address the temporal limitations by processing sequential data and retaining memory of prior states. Their ability to capture long-term dependencies makes them well-suited for analyzing emotional changes across time. Research shows that LSTM-based models can identify early indicators of rising stress and even predict stress escalation by analyzing sequences of emotional data [7]. This predictive capacity supports proactive intervention – for example, warning signs of accumulating stress can trigger coping prompts before symptoms worsen. However, LSTMs alone struggle with extracting rich spatial features from raw facial inputs. They often require pre-processed feature sequences (e.g., facial landmarks or features learned by a CNN), which limits real-world applicability where end-to-end feature learning is desired [13].

1.2.4 Hybrid CNN-LSTM Architectures

Given the complementary strengths and weaknesses of CNNs and LSTMs, recent literature has focused on hybrid CNN-LSTM architectures. A CNN-LSTM approach integrates CNN-based spatial feature extraction with LSTM-based temporal modeling to capture both facial details and their changes over tim [9]. In such systems, the CNN component first learns facial expression features (e.g., muscle movements or Action Units) from each video frame, and those features are then fed into the LSTM which learns the temporal dynamics across frames [10]. This combined architecture can detect subtle stress-related expressions and also track their progression (or fluctuation) over time, yielding more accurate and robust stress detection than either model alone.

1.2.5 Edge Computing for Privacy-Preserving Real-Time Processing

Edge computing has become crucial in mental health and affective computing applications, where privacy protection and real-time responsiveness are essential. Recent studies demonstrate that emotion recognition algorithms can be deployed locally on smartphones, microcontrollers, or workplace devices, ensuring that sensitive video data never leaves the device [13]. Performing AI inference on the edge dramatically reduces latency and enables immediate feedback, which is vital for timely stress interventions in fast-paced work environments [14]. By eliminating the need to transmit facial images or biometric data to external cloud servers, on-device processing directly addresses privacy and security concerns that often hinder adoption of monitoring technologies. Importantly, several works have achieved on-device emotion recognition with accuracy comparable to cloud-based models by optimizing and compressing network architectures [10]. The trade-off, however, is that resource-constrained devices limit model size and complexity, forcing developers to balance accuracy with efficiency. This necessitates optimized, lightweight models to ensure high performance within device limitations [12].

1.2.6 Personalized Intervention Systems and Alert Mechanisms

Research on personalized stress intervention systems remains limited, highlighting a significant gap in current stress management technology. Most existing digital stress management tools apply generic, one-size-fits-all coping suggestions (e.g., a standard relaxation or breathing exercise) without accounting for individual effectiveness, context, or predictive indicators of stress [2]. These rule-based responses ignore factors like personal stress triggers, historical coping success, or the user's current environment. While some emerging systems have begun to explore just-in-time adaptive interventions (using sensor data and AI to tailor the timing or content of support), such approaches are still in early stages [8]. Few workplace platforms integrate a truly adaptive recommendation engine that analyzes each employee's stress history, real-time physiological/emotional data, and situational context to automatically select the optimal intervention [7], [9]. Furthermore, most systems operate on scheduled check-ins or user-initiated help requests, often sending notifications after stress has peaked rather than delivering real-time, preventative alerts [14]. The lack of immediate, context-aware personalization in today's solutions highlights a major unmet need in stress technology [15].

1.3 Research Gap

Stress detection and mental health monitoring in workplace environments present unique challenges that current technologies have not fully addressed, particularly in real-time, privacy-preserving settings. Traditional approaches, which primarily rely on self-reporting mechanisms and physiological sensors, are reactive in nature and often miss critical opportunities for early intervention. Although advances in computer vision and machine learning have led to the development of AI-based emotion recognition systems, these solutions are not without significant limitations.

Recent research initiatives have demonstrated how deep learning techniques can be applied to facial expression recognition for stress detection. However, these approaches are designed for quite specific use cases and do not address the broader requirement for comprehensive stress management systems that can operate in real-world workplace environments while providing immediate, personalized interventions. Furthermore, while individual deep learning architectures perform well in controlled situations, they frequently lose effectiveness when subjected to the complex temporal dynamics and privacy constraints found in practical applications.

Moreover, the majority of existing emotion recognition systems lack the integration necessary for real-time stress detection coupled with intelligent intervention delivery. This gap emphasizes the necessity for an innovative solution that leverages the latest developments in hybrid deep learning architectures, edge computing, and personalized recommendation systems while adapting to the unique demands of workplace mental health monitoring. Such a system would enhance the accuracy and effectiveness of stress detection while providing immediate, context-aware support to prevent stress escalation.

Comparison with Existing Research

Research A [4]: Focuses on deep CNN models for stress detection through facial expression recognition, achieving high accuracy in spatial feature extraction but lacking temporal understanding of stress progression patterns and personalized intervention capabilities.

Research B [5]: Addresses real-time stress detection using integrated facial expression analysis with hybrid neural networks, successfully combining CNN with other methods but limited compared to CNN-LSTM spatio-temporal modeling and without intelligent intervention selection.

Research C [9]: Implements real-time stress detection using facial images with CNNs, demonstrating edge computing capabilities for privacy preservation but limited to simple CNN architecture without temporal analysis or personalized recommendation systems.

Research D [12]: Develops CNN combined with Haar cascade techniques for real-time emotion-based stress detection in AI systems, achieving effective detection capabilities but lacking temporal modeling and comprehensive intervention delivery mechanisms.

Research E [14]: Explores facial expression recognition systems using transfer learning approaches (VGG16, VGG19, Inception-ResNet V2) for stress detection, providing strong benchmark performance but without hybrid temporal analysis or dynamic intervention selection capabilities.

Table 1 : Research Gap

Research	**CNN-Based Spatial Analysis**	**LSTM-Based Temporal Analysis**	**Hybrid Architecture**	**Edge Computing Privacy**	**Personalized Intervention Selection**
Research A: Deep CNN Stress Detection [4]	✓	✗	✗	✗	✗
Research B: Hybrid Neural Network [5]	✓	✗	*Limited	✗	✗
Research C: Real-Time CNN Edge Detection [9]	✓	✗	✗	✓	✗
Research D: CNN-Haar Cascade System [12]	✓	✗	✗	✗	✗
Research E: Transfer Learning Approach [14]	✓	✗	✗	✗	✗
Proposed System	✓	✓	✓	✓	✓

*Limited = Basic hybrid approach without CNN-LSTM spatio-temporal modeling

Most of the existing research correctly emphasizes improving detection accuracy and real-time processing capabilities. However, critical areas such as hybrid architectural integration, dynamic intervention selection based on predictive analysis, comprehensive personalization engines, and the seamless integration of detection-to-intervention workflows haven't received adequate attention in the research mentioned above.

The lack of hybrid CNN-LSTM integration represents a fundamental limitation, as no existing system [4, 5, 9, 12, 14] combines CNN spatial feature extraction with LSTM temporal analysis for comprehensive emotion understanding. While Research A [4] and Research E [14] demonstrate strong CNN-based spatial analysis, and Research B [5] attempts hybrid approaches,

none achieve true spatio-temporal modeling that can capture both immediate facial expressions and stress progression patterns over time.

Furthermore, the absence of dynamic intervention selection is evident across all reviewed systems, where current approaches provide generic recommendations without intelligent algorithms that can predict intervention effectiveness based on individual patterns and contextual factors. Research D [12] and Research C [9] focus solely on detection without any intervention capabilities, while existing intervention systems lack sophisticated recommendation engines that can process multiple intervention options and deliver personalized suggestions within seconds of stress detection.

The incomplete integration workflows problem is particularly apparent, as existing research focuses on individual components (detection OR intervention) rather than seamless end-to-end stress management ecosystems. Research A [4], Research C [9], and Research E [14] excel in detection accuracy but provide no intervention mechanisms, creating a critical gap between stress identification and practical user support.

Additionally, privacy-performance trade-offs remain unresolved, as current edge computing implementations like Research C [9] sacrifice detection accuracy for privacy preservation, without optimized architectures that can maintain both high performance and complete data privacy through advanced hybrid modeling approaches.

Neglecting these integration challenges could lead to systems that detect stress effectively but fail to provide meaningful, timely support when users need it most. Therefore, a comprehensive approach should combine spatial and temporal analysis capabilities with intelligent intervention selection and real-time delivery mechanisms to ensure the derived stress management outcomes are trustworthy, effective, and practically applicable in real workplace environments.

1.4 Research Problem

Work-related stress is one of the most pressing challenges in modern professional environments, significantly impacting employee health, productivity, and organizational sustainability. Traditional stress detection methods, such as self-report surveys and periodic assessments, are reactive, subjective, and incapable of capturing real-time emotional fluctuations. As a result, opportunities for early intervention are often missed, leading to escalating stress, burnout, and reduced workplace performance.

Recent advances in artificial intelligence (AI), computer vision, and edge computing provide a pathway toward proactive, real-time stress monitoring systems that can detect early signs of stress and deliver personalized interventions. However, existing approaches remain limited in four critical ways: (1) over-reliance on static CNN models that ignore temporal dynamics of stress progression, (2) lack of personalization to account for individual emotional variability, (3) reactive rather than proactive intervention mechanisms, and (4) unresolved privacy concerns arising from cloud-based data processing.

First, most current systems rely heavily on CNN-based spatial analysis, which can detect facial features and micro-expressions but fail to capture the temporal evolution of stress. Stress is not static, develops through subtle, progressive changes over time. Without modeling these dynamics, systems cannot anticipate escalating stress levels. While some hybrid models exist, they remain limited and do not achieve comprehensive CNN–LSTM integration that could provide robust spatio-temporal emotion recognition.

Second, there is a lack of personalization in existing frameworks. Current systems adopt a one-size-fits-all model, ignoring the fact that stress expression varies with personality, cultural background, and baseline emotional state. This leads to frequent false positives and negatives. Moreover, interventions are typically generic and static, restricted to reminders or relaxation prompts, without adapting dynamically to user history, preferences, or current context. The absence of intelligent recommendation engines capable of selecting the most effective, context-aware intervention in real time remains a major gap.

Third, most stress detection systems are reactive rather than proactive. They identify stress only after it has escalated, offering limited coping strategies that fail to prevent stress from impacting performance. The detection-to-intervention gap persists systems may detect stress accurately but rarely connect this detection to timely, adaptive interventions that support employees at the exact moment of need.

Finally, privacy and trust issues remain unresolved. Many systems depend on cloud-based processing or continuous video streaming, raising concerns about surveillance, data misuse, and employee acceptance. While edge computing has been proposed as a solution, most implementations are oversimplified, sacrificing accuracy for speed. Building optimized privacy-preserving architectures that combine real-time performance with strong data protection is still an open challenge.

In summary, while existing research [4], [5], [9], [12], [14] has advanced stress detection through CNNs, hybrid models, and transfer learning, the field remains fragmented, with solutions focusing on either detection accuracy, real-time processing, or privacy—but rarely all together. The core research problem addressed in this study is the design and implementation of an AI-powered, real-time stress detection and intervention system that unifies spatio-temporal emotion recognition, personalized adaptive interventions, proactive stress management, and privacy-preserving edge deployment. This comprehensive framework seeks to bridge the gap between detection and action, enabling employees to receive the right support at the right moment.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to design, develop, and evaluate an AI-powered real-time stress detection and intervention system for workplace environments that provides accurate, privacy-preserving, and context-aware support for employees. Unlike traditional stress

management approaches that are reactive, subjective, and generic, the proposed system aims to establish a comprehensive closed-loop framework that continuously monitors employees' emotional states using hybrid Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models, captures both spatial and temporal features of facial expressions to analyze stress progression, and delivers personalized, just-in-time adaptive interventions. The system integrates edge computing architectures to ensure low-latency, real-time operation while safeguarding user privacy by processing data locally. By bridging the gap between stress detection and actionable intervention, this research seeks to transform workplace stress management from a reactive paradigm into a proactive, intelligent, and user-centered ecosystem that improves employee well-being, enhances productivity, and reduces the risk of burnout.

2.2 Specific Objectives

The proposed research is guided by several specific objectives that collectively ensure the development of a robust, intelligent, and practical stress detection and intervention system for workplace environments.

1. Develop a hybrid CNN–LSTM stress detection model.
The first objective is to construct and train a deep learning architecture that integrates Convolutional Neural Networks (CNNs) for spatial feature extraction with Long Short-Term Memory (LSTM) networks for temporal sequence modeling. CNNs will capture fine-grained spatial cues such as micro-expressions and facial muscle movements, while LSTMs will analyze how these expressions evolve over time to detect the buildup of stress. This dual modeling approach addresses the limitations of CNN-only static analysis and LSTM-only sequential analysis, enabling more accurate detection of subtle and progressive stress signals. The proposed model will be benchmarked against baseline CNN and LSTM systems to demonstrate improvements in spatio-temporal accuracy.

2. Implement a privacy-preserving edge computing framework.
Another key objective is to ensure that the stress detection system operates in real time while preserving user privacy. To achieve this, the system will be deployed on local edge devices such as workplace PCs, laptops, or on-premises servers, allowing all computations to occur locally without transmitting sensitive facial or biometric data to external cloud servers. By combining optimized lightweight models with edge computing capabilities, the system will reduce latency and enhance responsiveness while simultaneously fostering employee trust by protecting their personal information.

3. Design intelligent recommendation and intervention engines.
Beyond detection, the system must actively support employees in regulating stress. To achieve this, an adaptive recommendation engine will be developed that selects interventions from a diverse pool of evidence-based strategies, including guided breathing exercises, mindfulness prompts, calming music, or short break reminders. The selection will not be random or rule-based but will instead rely on predictive analysis, using historical user data, contextual

information, and stress severity levels to dynamically recommend the most effective intervention for each situation.

4. Enhance personalization and adaptability of interventions.
A major limitation of current stress management systems is their reliance on generic interventions. This research addresses that by incorporating personalization models capable of learning each user's unique stress response patterns over time. Factors such as personal coping mechanisms, cultural background, and individual preferences will guide intervention selection, ensuring recommendations are perceived as supportive rather than intrusive. For example, the system will avoid interrupting an employee during meetings and instead deliver context-aware prompts at appropriate times, fostering acceptance and effectiveness of the interventions.

5. Validate system robustness in real-world workplace conditions.
To ensure practical applicability, the system will be rigorously tested in real workplace settings where environmental factors such as varying lighting conditions, changing camera angles, multitasking activities, and demographic diversity can challenge recognition accuracy. Unlike models evaluated solely in controlled laboratory conditions, this research emphasizes resilience in unpredictable office environments. Successful validation under such conditions will demonstrate the system's reliability and scalability for widespread workplace deployment.

6. Assess user trust, usability, and adoption potential.
Finally, this research aims to investigate employee perceptions of the system's transparency, usefulness, and privacy safeguards. Trust is critical for adoption, and employees must feel confident that their data is secure and not being misused for surveillance. To achieve this, unobtrusive feedback mechanisms, clear communication about how data is handled, and user-friendly interfaces will be incorporated. By prioritizing usability and transparency, the objective is to ensure long-term adoption of the system, making it a sustainable tool for workplace stress management rather than a short-lived pilot project.

3. METHODOLOGY

This research study employs a comprehensive methodology that leverages cutting-edge deep learning, computer vision, and edge computing techniques to develop an AI-powered real-time stress detection and intervention system for workplace environments. To achieve the research objectives, a systematic methodological approach will be adopted, encompassing the following key phases:

3.1 System Overview

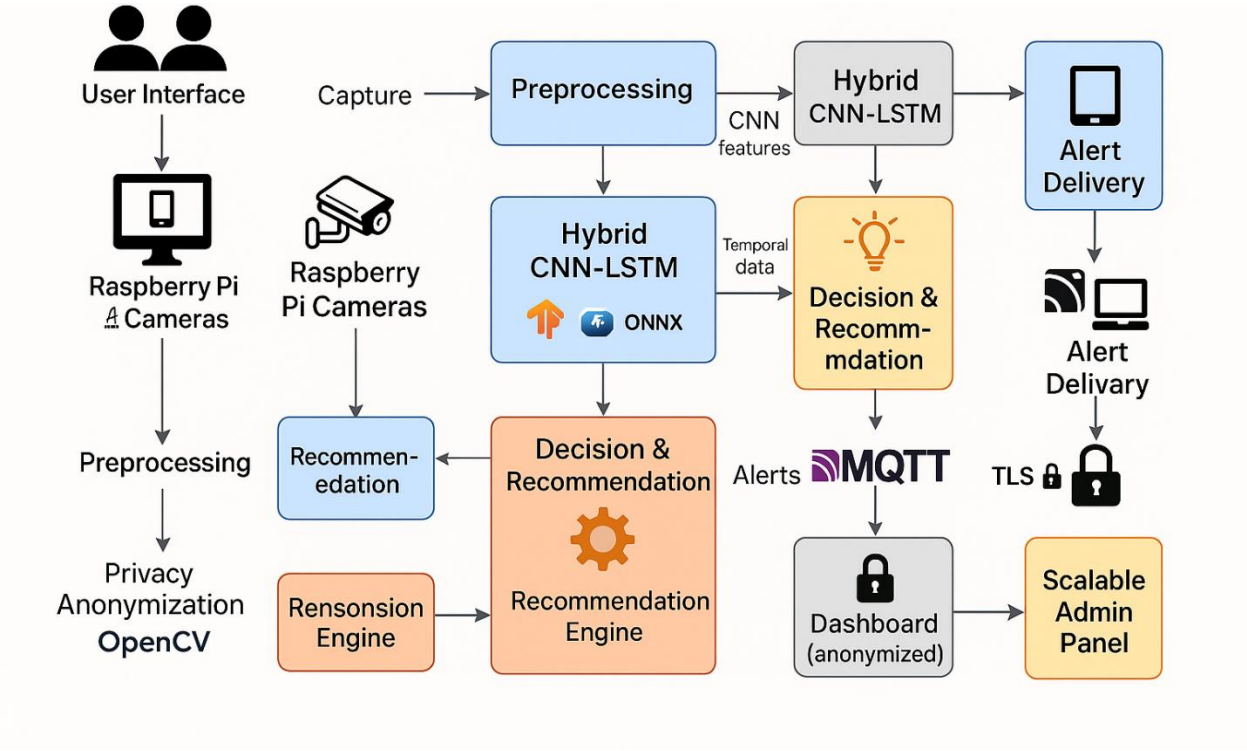

Figure 1 : System Overview

3.2 Requirements Gathering and Analysis

3.2.1 Literature Review

A comprehensive review of existing research on stress detection, emotion recognition, and AI-driven workplace well-being systems [4, 5, 9, 12, 14] will be conducted. This review helps to identify technical requirements, current limitations (e.g., lack of temporal modeling and personalization), and performance benchmarks. Insights from literature also guide the selection of suitable models (CNNs, LSTMs), preprocessing methods, and privacy-preserving mechanisms that align with workplace deployment needs.

3.2.2 Industry Consultations

Structured interviews and focus group discussions will be held with senior engineers in the IT industry aged 30–40, along with HR managers and occupational psychologists. This demographic is chosen because mid-career IT professionals face heavy workloads, high responsibility, and dual personal-professional commitments, making them highly vulnerable to stress. Consultations will gather insights on workplace stress patterns, preferred intervention

methods (e.g., short breaks, breathing prompts, mindfulness), and major concerns about privacy and surveillance.

3.2.3 Practical Observation

Direct workplace observation will be conducted in IT environments to understand real-world conditions such as lighting, camera placement, employee interactions, and task intensity. This ensures the system accounts for natural variability (e.g., multitasking, diverse postures, variable office lighting). Observations will also help validate whether interventions are context-appropriate and non-disruptive during critical work moments like meetings or coding sessions.

3.2.4 Regulatory and Ethical Requirements

The study will evaluate compliance with workplace privacy regulations, IT industry data-handling policies, and ethical guidelines for employee monitoring. Special emphasis will be placed on maintaining employee trust by ensuring local processing (edge computing), data anonymization, and consent-based participation. These requirements guarantee that the proposed system aligns with both legal frameworks and ethical standards in professional IT settings.

3.3 Data Collection

The data collection process for this research is designed to ensure ecological validity, reliability, and privacy preservation while capturing stress-related facial expressions and temporal dynamics in real workplace environments. Data will be collected entirely from employees using Raspberry Pi 4B units equipped with 8MP camera modules, ensuring that all processing and storage remain local to maintain privacy.

Data Collection Strategy

Data collection in this research focuses entirely on creating a custom dataset from real workplace environments using Raspberry Pi 4B units equipped with 8MP cameras.

- **Facial Expression Data**: Employees will be recorded during regular work activities under varied conditions (lighting changes, camera angles up to 30°, and different office settings). This ensures the dataset reflects natural, real-world variability rather than controlled laboratory conditions.
- **Temporal Sequence Data**: Longer recording sessions (45–60 minutes) will be conducted to capture full stress episodes, from onset to escalation and recovery. These sessions will be segmented into 30–60 second clips at 20 fps, allowing analysis of gradual build-up and acute spikes in stress.
- **Intervention Response Data**: During and after system-generated interventions (e.g., breathing prompts, mindfulness cues, or break reminders), facial expressions and

behavioral responses will be recorded. This enables the creation of an intervention-effectiveness dataset that links stress cues with adaptive coping strategies.

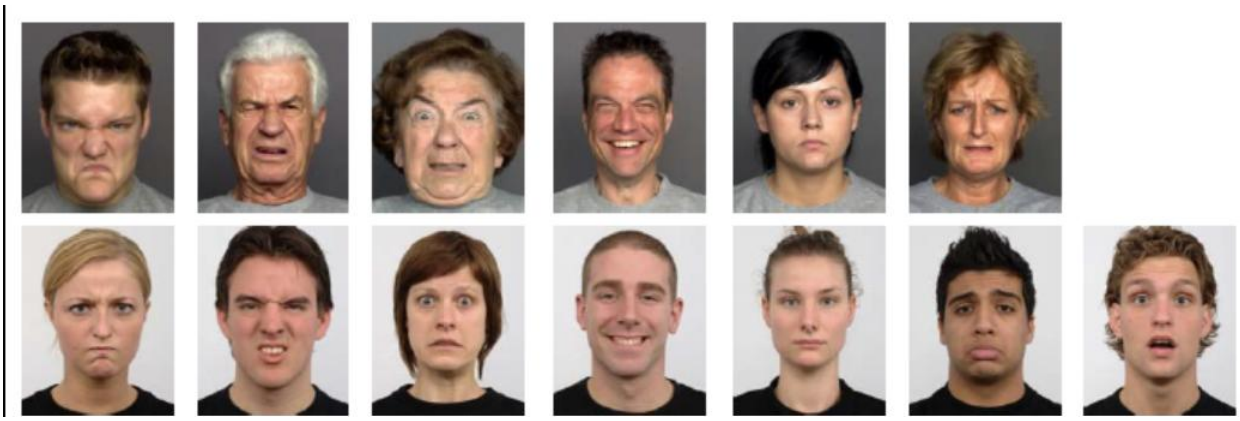

Figure 2 : Emotions of data collection [15]

3.4 Data Preprocessing

3.4.1 Data Preprocessing

Face Detection and Alignment

Faces are detected using OpenCV Haar cascade classifiers, which are optimized for Raspberry Pi's computational constraints. Once detected, dlib's 68-point landmark predictor is applied for geometric normalization of eyes, nose, and mouth, ensuring consistent alignment. GPU acceleration via Raspberry Pi's VideoCore processor (or optional accelerators such as Intel Movidius) supports real-time efficiency.

Image Enhancement

OpenCV-based preprocessing is applied to improve image quality under varied workplace conditions. Adaptive histogram equalization (CLAHE) corrects illumination variations, bicubic interpolation resizes images to 224×224 pixels, and Gaussian filtering reduces sensor noise while preserving micro-expression details such as wrinkles and furrows.

Temporal Sequence Preparation

Using OpenCV's video processing utilities, video streams are standardized to 20 fps and segmented into overlapping sliding windows. When face detection fails, missing frames are interpolated with optical-flow estimation to maintain temporal continuity. This ensures smooth sequences for LSTM training.

Privacy Preprocessing

All preprocessing is performed locally on Raspberry Pi devices. OpenCV and Python-based pipelines anonymize outputs by removing identity-specific features while preserving stress-

relevant cues. Processed data are encrypted (AES-256), stored locally with timed auto-deletion, and differential-privacy noise is added to sensitive features before model training.

Dataset Splitting
To support training and evaluation, the final dataset is divided into training (70%), validation (15%), and testing (15%) subsets. This ensures robust performance evaluation and fair model comparisons.

3.4.2 Quality Assurance

Annotation Control
Expert annotators, including psychologists, assign stress labels with an inter-rater agreement threshold of $\kappa \geq 0.85$. Temporal continuity checks are performed to avoid inconsistent labeling, and where possible, annotations are cross validated against self-reported stress levels or physiological measures.

Dataset Validation
Systematic checks are carried out to ensure balanced stress distributions (30% low, 40% moderate, 30% high), demographic representation, and dataset integrity. Pilot CNN–LSTM training is used as a benchmark test to confirm annotation quality and dataset usability. Iterative refinement continues until the dataset consistently supports robust stress detection and intervention modeling.

3.5 Hybrid CNN-LSTM Architecture Development

3.5.1 CNN Component Design

Spatial Feature Extraction Architecture:

- Base CNN architecture: Modified ResNet-50 for facial feature extraction
- Custom convolutional layers for micro-expression detection
- Attention mechanisms for focusing on stress-relevant facial regions
- Feature map dimension: 2048 features per frame

3.5.2 LSTM Component Design

Temporal Pattern Analysis Architecture:

- Bidirectional LSTM with 256 hidden units
- Attention-based temporal modeling for stress progression analysis
- Dropout regularization (0.3) to prevent overfitting
- Sequence length: 30 frames (approximately 5-10 seconds)

3.5.3 Hybrid Integration Strategy

- CNN feature extraction followed by LSTM temporal modeling
- Feature fusion techniques for combining spatial and temporal information
- Multi-task learning for stress level classification and progression prediction
- Model optimization for edge deployment constraints

3.6 Intelligent Recommendation Engine Development

3.6.1 Intervention Pool Creation

A structured intervention pool will be developed, covering five evidence-based categories: breathing exercises (4-7-8, box breathing, diaphragmatic breathing), mindfulness prompts (body scan, gratitude practice, present-moment awareness), music therapy (classical, nature sounds, binaural beats), physical interventions (stretching, posture correction, short breaks), and cognitive techniques (positive affirmations, reframing exercises). This diversity ensures flexibility and scientific grounding to support varied stress profiles.

3.6.2 Recommendation Algorithm Design

The recommendation algorithm will use a multi-factor decision framework that considers stress severity (low, moderate, high), historical intervention effectiveness, contextual factors such as workload or meeting schedules, user preferences and feedback history, and intervention characteristics like duration and level of disruption. This ensures interventions are context-aware, minimally disruptive, and optimally matched to the employee's current situation.

3.6.3 Personalization Engine

Personalization will be achieved through a combination of collaborative filtering to identify effective strategies across similar users, content-based filtering to match interventions with individual coping styles, reinforcement learning to improve recommendations dynamically through real-time feedback, and A/B testing frameworks to evaluate alternative strategies. Over time, this engine evolves into an adaptive, intelligent system that delivers increasingly personalized and effective interventions.

3.7 Real-Time Communication and Alert System

3.7.1 Communication Protocol Implementation

The system will employ a lightweight MQTT-based architecture for transmitting real-time stress detection outputs. Stress-level updates will be published to topic-based channels (e.g., low, moderate, high stress), allowing targeted delivery. Quality of Service (QoS) level 1 will ensure guaranteed message delivery, while TLS encryption secures all communications, preserving data confidentiality during transmission between Raspberry Pi devices and end-user applications.

3.7.2 Alert Delivery Mechanisms

Alerts will be designed to maximize support without disrupting productivity. Delivery mechanisms include:

- Push notifications to mobile and desktop applications for immediate awareness.
- Escalation protocols, where persistent high-stress levels trigger additional interventions or notify support personnel (in anonymized form).
- Context-aware delivery, ensuring alerts are timed to avoid critical activities like scheduled meetings or coding sessions.
- Multi-modal notifications (visual pop-ups, gentle audio cues, or haptic feedback for wearables), giving employees flexibility to choose their preferred method.

3.8 Dashboard Visualization

The system includes a dashboard that automatically records each stress detection event and corresponding intervention. For every instance, the dashboard logs the time of detection, the stress level, the suggested intervention, and any feedback or outcome. This history view allows employees to monitor their own stress patterns over time, while organizations (with anonymized access) can track broader trends. Visual elements such as time-series charts, intervention effectiveness summaries, and weekly reports make it easier to identify recurring stress periods and evaluate which strategies work best. Beyond simple logging, the dashboard serves as an adaptive insight tool, feeding back into the recommendation engine to improve future personalization.

3.9 Development Methodology

The development of the proposed system will follow an Agile SCRUM framework to ensure iterative progress, continuous feedback, and adaptability to real-world requirements. This approach allows close collaboration with stakeholders (employees, HR professionals, and occupational psychologists) while maintaining flexibility in refining system features.

3.9.1 Agile SCRUM Framework

The project will be structured into 2-week sprints, each delivering functional increments of the system.

- Daily stand-up meetings will be held for quick progress updates and issue resolution.
- Sprint planning and retrospectives will help define tasks, evaluate progress, and refine workflows.

- Continuous integration and deployment (CI/CD) pipelines will ensure seamless model updates and system integration.
- Regular stakeholder feedback will be incorporated, especially regarding usability, privacy, and intervention effectiveness, to keep the solution user-centered.

3.9.2 Development Phases

Model Development and Training (Months 1–4)

- Implementation of the CNN model for baseline feature extraction and later extension to a hybrid CNN–LSTM for spatio-temporal stress analysis.
- Data preprocessing (face alignment, enhancement, anonymization) and training using the custom workplace dataset.
- Edge optimization (e.g., lightweight model compression, quantization) for Raspberry Pi deployment with low latency and strict privacy.
- Initial model validation against annotated datasets to establish baseline accuracy.

System Integration and Testing (Months 5–8)

- Development of the recommendation engine using multi-factor decision criteria and reinforcement learning for personalization.
- Implementation of the communication framework using MQTT and secure TLS protocols for real-time stress alerts.
- Integration of the dashboard visualization module, enabling automated logging of stress events, interventions, and outcomes.
- End-to-end system testing in controlled workplace-like environments to validate functionality.

Field Testing and Validation (Months 9–12)

- Deployment of the system in real-world workplace environments with employees aged 30–40 (target demographic).
- User acceptance testing to assess trust, usability, and perceived privacy compliance.
- Collection of feedback on stress detection accuracy and intervention effectiveness.
- Final performance optimization and deployment, ensuring the system meets objectives of real-time operation, personalization, and privacy preservation.

3.10 Technologies and Tools

• **Programming Language:** Python
• **Frameworks:** TensorFlow/Keras, PyTorch, scikit-learn
• **Computer Vision Libraries:** OpenCV
• **Object Detection / Modeling:** CNN, LSTM, CNN–LSTM Hybrid, TensorFlow Lite, ONNX Runtime, NVIDIA TensorRT
• **IoT Platforms / Communication:** MQTT, REST APIs, WebSocket, Firebase, Docker, Node.js
• **Development Tools:** Jupyter Notebook, Google Colab, Visual Studio Code, Grafana, React Native
• **Hardware:** Raspberry Pi 4B with 8MP Camera Module, Workplace PCs/Laptops, Mobile Devices (Android/iOS)

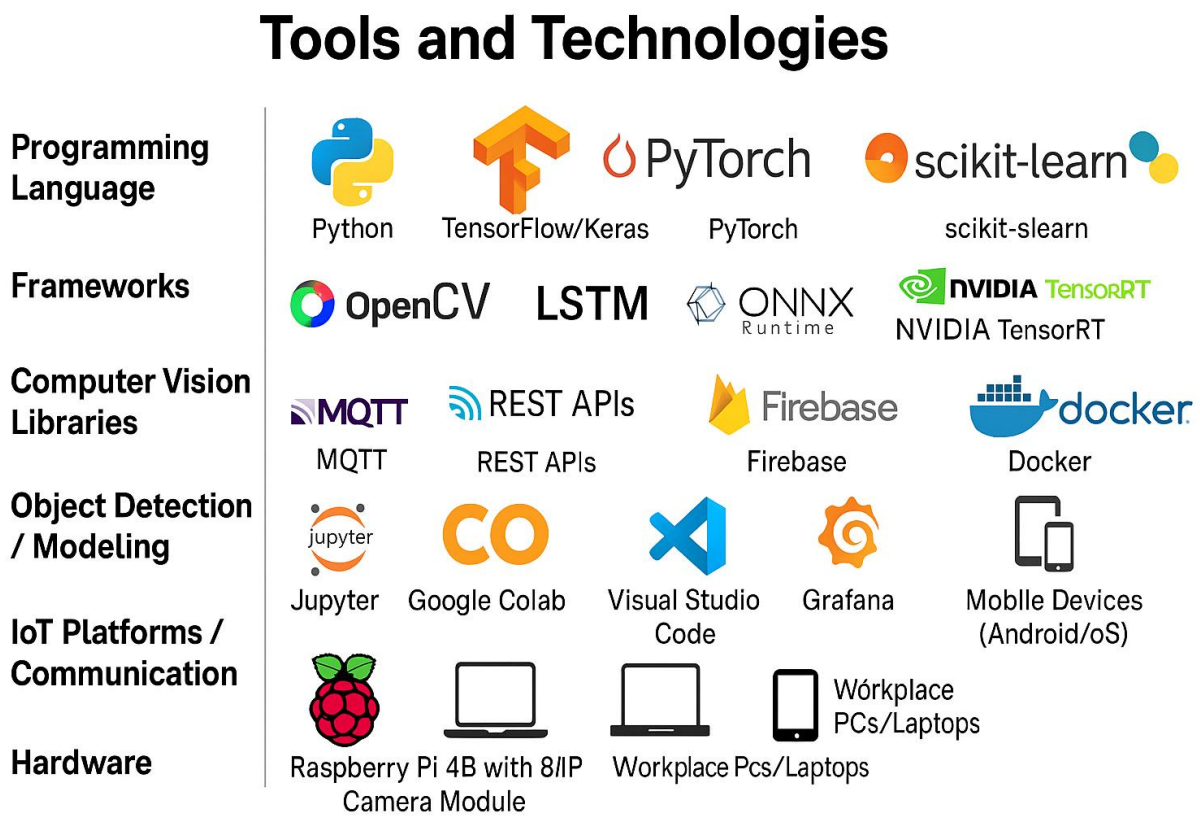

Figure 3 : Technologies and Tools

3.11 Feasibility Study

3.11.1 Technical Feasibility

The proposed research integrates proven technologies including CNN facial recognition, LSTM temporal analysis, edge computing deployment, and MQTT communication protocols. The development team possesses expertise in deep learning, computer vision, mobile application development, and edge computing optimization, ensuring technical viability of the integrated system.

3.11.2 Economic Feasibility

Development Costs:

- Software development and testing infrastructure
- Cloud computing resources for model training
- Edge device procurement for testing and deployment
- User study compensation and data collection expenses

The primarily software-based solution requires minimal hardware investment, making it economically viable for workplace deployment.

3.11.3 Schedule Feasibility

The 12-month development timeline accommodates comprehensive model development, system integration, real-world testing, and validation phases. The agile development methodology enables iterative improvement and stakeholder feedback incorporation throughout the process.

3.11.4 Legal and Ethical Feasibility

- Compliance with workplace privacy regulations and mental health data protection standards
- Informed consent procedures for data collection and system usage
- Ethical approval from institutional review boards
- Clear data ownership and usage agreements with participating organizations

3.12 System Development Methodology

The system development will follow an Agile SCRUM methodology with iterative development cycles, enabling continuous refinement based on user feedback, model evaluation, and real-world testing in workplace environments. Each sprint will deliver functional components that are integrated, tested, and optimized for edge deployment.

3.12.1 Stress Detection Module Development

- **Dataset Preparation**: Collection of facial expression data from employees (ages 30–40) using Raspberry Pi cameras under varied workplace conditions. Annotation of stress intensity levels (0–100 scale) and intervention-response labeling.
- **Model Training**: Development of CNN models for initial feature extraction, followed by integration into CNN–LSTM hybrids for spatio-temporal stress recognition.
- **Performance Optimization**: Conversion of trained models into lightweight TensorFlow Lite / ONNX formats, enabling efficient inference on Raspberry Pi and workplace PCs.

3.12.2 Edge Computing & IoT Integration

- **Raspberry Pi Deployment**: Integration of Raspberry Pi 4B with 8MP camera modules for on-device data capture, local preprocessing, and inference.
- **Communication Protocol Implementation**: Secure real-time data transfer using MQTT for stress alerts, REST APIs for system integration, and WebSocket for bidirectional updates.
- **Privacy Safeguards**: On-device anonymization, AES-256 encryption, and auto-deletion of raw data to ensure compliance with ethical and legal standards.

3.12.3 Recommendation & Personalization Engine

- **Intervention Pool Design**: Building a library of evidence-based interventions (e.g., breathing exercises, mindfulness prompts, music, short breaks).
- **Recommendation Algorithm**: Multi-factor decision-making considering stress severity, context, history, and user preferences.
- **Personalization Mechanisms**: Incorporating collaborative filtering, reinforcement learning, and feedback loops to adapt interventions dynamically over time.

3.12.4 Dashboard & Notification System

- **Dashboard Development**: Grafana-based visualization of stress history, intervention logs, and analytics, updated in real time from detection modules.
- **Notification Optimization**: Delivery of context-aware, multi-modal alerts (push notifications, desktop pop-ups, mobile prompts) to minimize disruption.

- **Integration Testing**: Comprehensive validation of alert accuracy, dashboard synchronization, and intervention records across mobile and desktop platforms.

3.13 Project Requirements

3.13.1 Functional Requirements

Real-Time Stress Detection
The system must continuously monitor employees' facial expressions in real time using Raspberry Pi cameras, achieving at least 90% accuracy in stress classification across low, moderate, and high levels.

Spatio-Temporal Analysis
The detection model must capture both spatial features (micro-expressions) and temporal patterns (stress progression) using a CNN–LSTM hybrid. Predictions must update within 1–2 seconds of new frame input.

Privacy-Preserving Edge Processing
All video data must be processed locally on Raspberry Pi devices or workplace PCs. No raw facial data should be transmitted externally, ensuring privacy compliance.

Adaptive Intervention Recommendations
The system must generate personalized interventions (e.g., breathing prompts, mindfulness cues, short breaks, or music therapy) within 5 seconds of detecting high stress. Recommendations must consider historical effectiveness, user preferences, and contextual factors.

Dashboard Logging and Visualization
The system must automatically update the dashboard with stress events, intervention history, and user feedback. Logs must include timestamps, stress levels, and chosen interventions, with weekly and monthly reports available for both users and organizations (in anonymized form).

Alert and Notification System
The system must deliver immediate alerts through multiple channels including desktop apps, mobile notifications, and dashboards. Alerts must be context-aware to avoid interruptions during critical tasks such as meetings.

Scalability and Adaptability
The framework must support deployment across multiple workplace setups, from small teams (5–10 employees) to larger offices (>100 employees), with dynamic addition/removal of monitoring devices.

3.13.2 Non-Functional Requirements

System Performance
The system must process real-time video streams at 20 fps per camera with end-to-end detection and recommendation latency not exceeding 200 ms on Raspberry Pi or workplace PCs.

Reliability and Availability
The system should maintain 99% uptime with fault-tolerant operation. Stress events must not be lost; data loss tolerance must be less than 0.5% under normal operating conditions.

Security and Privacy
All communication (MQTT, REST, WebSocket) must be encrypted using TLS. Role-based access control and audit logs must be implemented to track system usage. Data storage must follow AES-256 encryption standards with auto-deletion protocols.

Resource Efficiency
Models deployed on Raspberry Pi devices must be lightweight (TensorFlow Lite/ONNX optimized) to ensure inference within hardware limits. Devices must operate continuously without overheating or exceeding power constraints.

Usability and User Trust
The interface must be simple and non-intrusive. Employees must understand how data is handled, with transparent explanations of privacy safeguards. Feedback collection should be integrated into the workflow without causing extra stress.

Extensibility
The system should be designed modularly, allowing future integration with third-party HR systems, wearable devices, or advanced intervention libraries without major redesign.

3.14 Expected test cases

1) Test Case - Real-Time Stress Detection and Classification

Objective: Verify that the CNN–LSTM hybrid model accurately detects low, moderate, and high stress levels from facial expressions in real time.
Steps:

1. Record participants (ages 30–40) during varied workplace activities (coding, meetings, multitasking).
2. Annotate ground truth stress levels (low, moderate, high) using psychologist ratings.
3. Stream live video to Raspberry Pi and run local inference.
4. Introduce stress triggers (e.g., workload tasks, time pressure) and recovery phases (short breaks).

Expected Results:

- Stress levels are classified correctly within 1–2 seconds of frame input.
- Predictions remain stable without oscillations from lighting/posture variations.

Measurement:

- Accuracy ≥90%, precision/recall ≥0.90 per class, F1 ≥0.90.
- Latency ≤200 ms end-to-end per frame.
- Fairness: accuracy gap ≤5% across demographics.

2) Test Case - Temporal Stress Progression Tracking

Objective: Validate the system's ability to capture temporal changes (stress build-up and recovery) using LSTM-based sequence modeling.
Steps:

1. Run 30–60 second video clips where stress increases gradually.
2. Feed these sequences into the CNN–LSTM pipeline.
3. Introduce recovery interventions and monitor system response.

Expected Results:

- The system identifies early signs of stress escalation before the participant's peak stress moment.
- Stress decline after intervention is recognized accurately.

Measurement:

- AUC ≥0.92 for escalation prediction.
- Early detection lead time ≥5 s before observed peak.
- Sequence classification latency ≤80 ms per window.

3) Test Case - Personalized Intervention Recommendation

Objective: Ensure that the recommendation engine selects context-aware, effective interventions based on user history and real-time context.
Steps:

1. Configure user profiles with preferences and feedback logs.
2. Simulate high stress during a meeting, coding task, and break time.

3. Deliver interventions (silent breathing prompt, short mindfulness, longer relaxation exercise).

Expected Results:

- Interventions match context (e.g., silent cue in meeting, guided prompt during break).
- Recommendations adapt based on user feedback ("helpful" vs. "not helpful").

Measurement:

- Context compliance ≥95%.
- Intervention acceptance rate ≥80%.
- Personalization improvement: helpfulness score ↑ ≥20% over 2 weeks.

4) Test Case - Real-Time Alerts and Notifications

Objective: Confirm secure, reliable delivery of stress alerts to user devices via MQTT/WebSocket and Firebase push notifications.
Steps:

1. Generate 20 high-stress events and 10 moderate events.
2. Deliver alerts to mobile and desktop applications in real time.
3. Users acknowledge alerts and follow recommendations.

Expected Results:

- Alerts delivered with correct payload (stress level, time, suggested intervention).
- No duplicate or lost alerts.
- User acknowledgment and closure logged in the system.

Measurement:

- Delivery success ≥99%; duplicates ≤1%; dropped alerts = 0.
- Detection → notification latency ≤3 s (median), ≤6 s (p95).
- Acknowledgment within 60 s in ≥90% cases.

5) Test Case - Dashboard Logging and Analytics

Objective: Validate that the dashboard accurately records stress events, interventions, and outcomes for both users and organizational overview.
Steps:

1. Simulate multiple stress detection and intervention cycles across one week.
2. Cross-check dashboard logs against raw system events.
3. Review historical analytics (time-series charts, effectiveness reports).

Expected Results:

- Logs include timestamp, stress level, chosen intervention, and feedback.
- Weekly and monthly summaries are generated automatically.

Measurement:

- Log accuracy ≥99% (match with raw events).
- Report generation time ≤30 s.
- Usability score ≥80/100 (SUS survey).

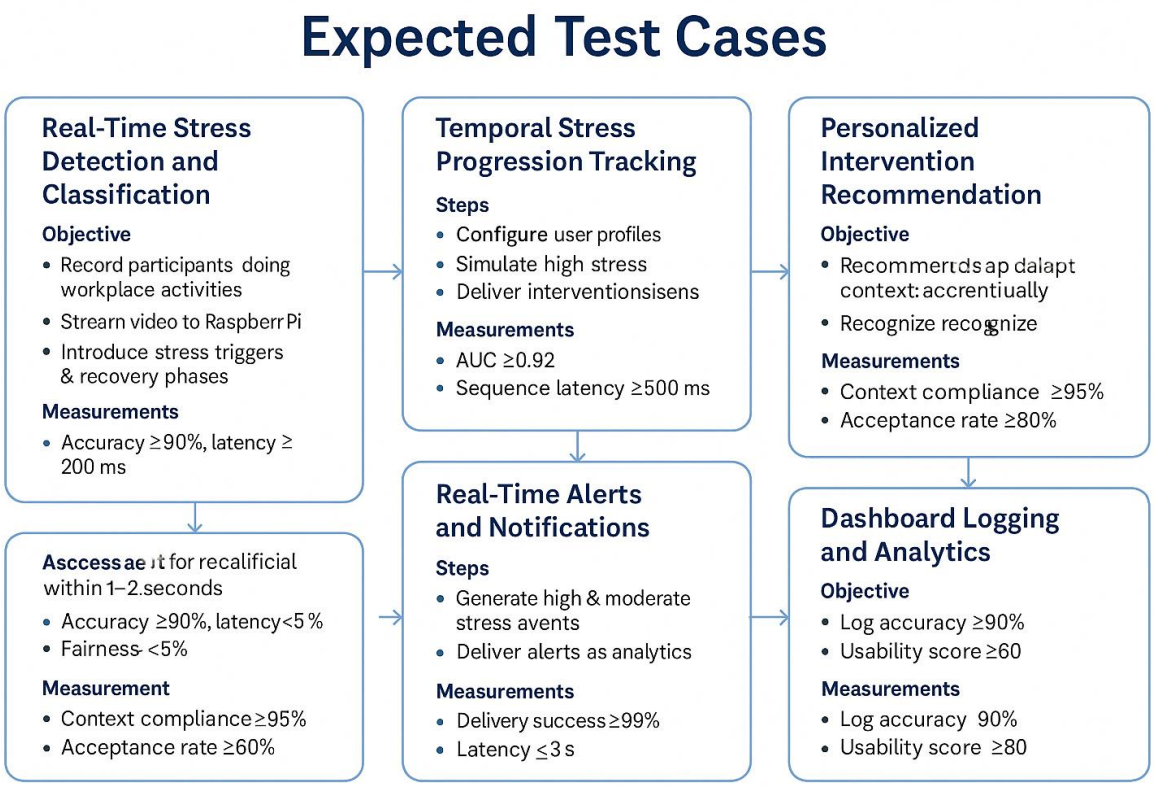

Figure 4 : Expected test cases

3.14 Work Breakdown Structure (WBS)

The following Work Breakdown Structure (WBS) illustrates the major phases and tasks involved in developing the proposed AI-powered real-time stress detection and intervention system for workplace environments. It provides a high-level overview of the project workflow, beginning with dataset preparation and preprocessing, followed by hybrid CNN–LSTM model development, edge computing and IoT integration, recommendation engine and personalization design, dashboard visualization, and finally testing and validation in real workplace environments. Each phase is broken down into key activities to ensure a structured and systematic approach to achieving the project objectives. This WBS serves as a roadmap for the research, ensuring clarity in task allocation, milestone tracking, and timely project completion while maintaining alignment with privacy, usability, and performance requirements.

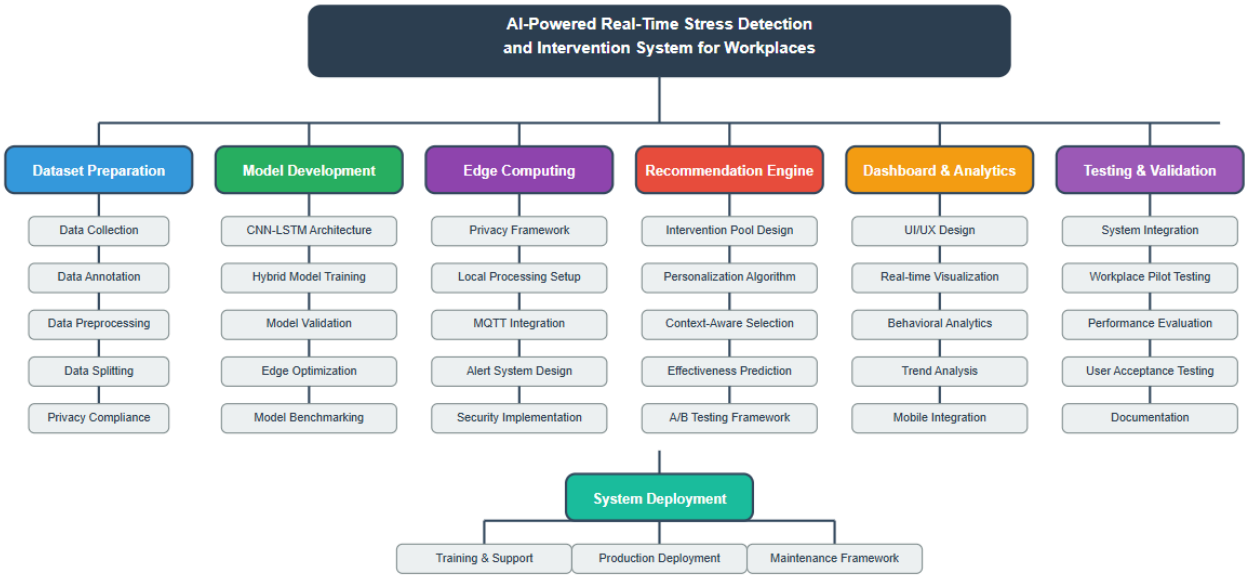

Figure 5 : Work Breakdown Structure

3.15 Gantt Chart

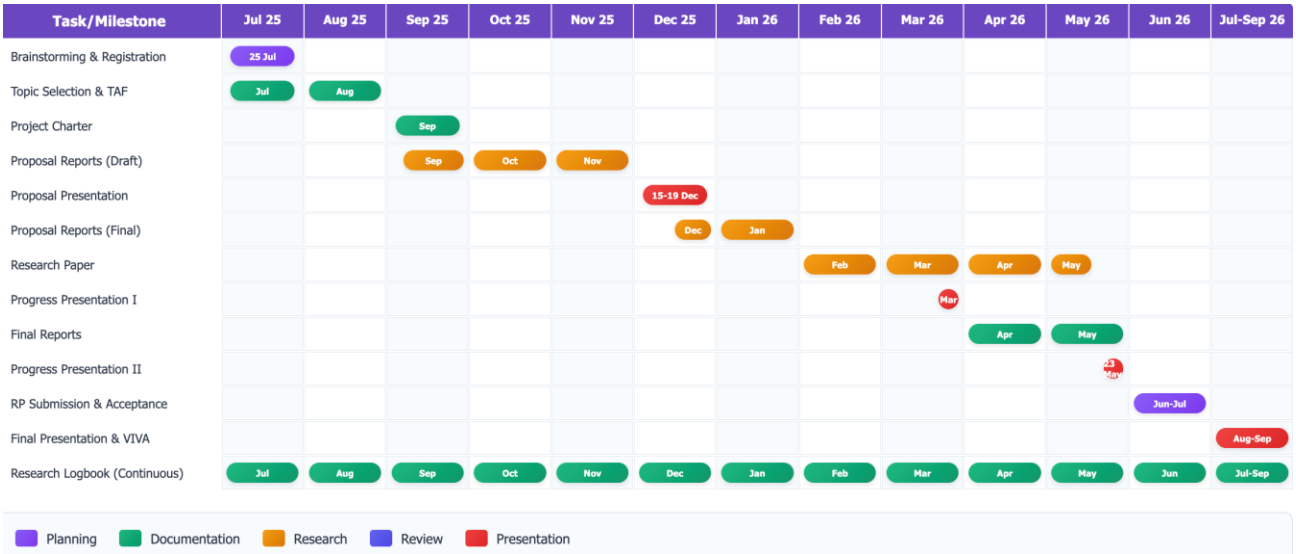

Figure 6 : Gantt Chart

The proposed research study will be executed within a structured 12-month timeline to ensure timely completion and systematic progress. The planning phase will commence in May 2025 and continue until July 2025, covering literature review, stakeholder consultations with IT engineers and psychologists, and requirement analysis. The project initiation phase will run from July 2025 to September 2025, focusing on finalizing the research proposal, defining system specifications, and preparing the custom dataset collection strategy using Raspberry Pi cameras. The implementation phase will begin in September 2025 and extend to February 2026, during which core system modules will be developed, including the hybrid CNN–LSTM model, privacy-preserving edge framework, and intelligent recommendation engine. This will be followed by the testing and evaluation phase from February to April 2026, involving controlled testing, workplace pilot deployment, user acceptance testing, and performance validation. Finally, the documentation and conclusion phase will take place between April and June 2026, covering the preparation of the full research paper, final system documentation, and the project defense presentation. This timeline ensures the research objectives are met systematically, aligning with both academic deadlines and real-world applicability.

4. DESCRIPTION OF PERSONNEL & FACILITIES

This research is supervised by Prof. Anuradha Jayakody, Professor at the Faculty of Computing, Sri Lanka Institute of Information Technology (SLIIT), Malabe, Sri Lanka. She provides high-level academic guidance, ensures research quality, and offers continuous support to align the study with scientific and ethical standards. The project is co-supervised by Mr. Ashvinda Iddamalgoda, who contributes technical expertise in IoT systems, computer networks, and cybersecurity.

The research is carried out by a group of four undergraduate students from the Department of Information Technology, Faculty of Computing, SLIIT. Each member contributes to a distinct subsystem under the umbrella topic "Emotix: a wearable AI system for employee assistant and mental wellness." The distribution of work is as follows:

1. Kariyawasam K.G.A.N (IT22927170) – AI-powered real-time stress detection and intervention system for employees using facial expressions, Leads the development of a real-time stress detection system using hybrid CNN-LSTM models for facial expression analysis. The system sends instant alerts with personalized interventions (e.g., breathing exercises, music, or break reminders) and visualizes stress trends on an interactive dashboard.

2. Perera L.W.D.K (IT22601810) – Responsible for developing a workplace assistant that combines biomedical and eye-tracking signals with environmental sensing, integrated with an AI chatbot and voice assistant to provide real-time adaptive feedback and improve employee productivity.

3. **Rathnayaka R.M.H.D (IT22560476)** – Focuses on a self-learning adaptive routine monitoring system using TinyML to learn employee daily patterns, detect deviations, and deliver personalized wellness reminders, adapting routines with employee confirmation for privacy and personalization.

4. Jayarathna S.A.V.P IT22560612 – Designs a secure IoT network communication framework with end-to-end encryption, VPN tunneling, and TPM 2.0 integration. A novel fake robot entry point is introduced to attract attackers, analyze behavior, and trigger AI-driven adaptive defenses for enhanced security.

5. BUDGET AND BUDGET JUSTIFICATION

The budget for this project is carefully allocated to support the successful implementation of the proposed stress detection and intervention system. The primary investment is in Raspberry Pi 4B units with 8MP camera modules (Rs. 5,500 each), which will serve as edge devices for real-time video capture, preprocessing, and local inference. These devices are critical for ensuring privacy-preserving data collection and real-time monitoring in workplace environments.

Additionally, workplace PCs/laptops will be leveraged for training and testing, while cloud services such as Google Colab or equivalent compute support will be used for advanced model training and fine-tuning during the research phase. To facilitate real-time communication and notifications, Firebase services (Rs. 1,000) will be utilized for push notifications and temporary data synchronization between devices and mobile applications.

This allocation ensures a cost-effective setup that balances hardware for edge deployment with cloud-based computational resources for training. The budget prioritizes scalability, real-time responsiveness, and strict privacy preservation.

Budget Breakdown

Table 2 : Budget Justification

Resource Type	**Unit Cost (LKR)**	**Quantity**	**Total (LKR)**
Raspberry Pi 4B + 8MP Camera Module	Rs. 5,500	1	Rs.15,500
Firebase Services (Push Notifications, DB)	Rs. 1,000	1	Rs. 1,000
Miscellaneous (Cables, Mounts, Power Supply, Storage)	Rs.500	1	Rs. 500
Total			**Rs.17,000**

This budget is justified as it ensures real-time stress monitoring, privacy-preserving edge deployment, and scalable cloud training support while remaining cost-efficient and practical for workplace testing.

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